

Karnataka Joint Gravity–Magnetic Inversion: Comparing Couplings

The gravity and the magnetic data of the same area inverted together on one mesh with terrain, once for each coupling of the joint inversion. Each model keeps the regularization of its single inversion; only the coupling differs. The report shows what each coupling does to the two models, what it costs in data fit, and which assumption suits this belt.

AREA	DATA
70 × 70 km, easting 641–711 km, northing 1634–1704 km (UTM 43N, EPSG:32643)	Complete Bouguer anomaly, 5,040 points, error 0.5 mGal; magnetic anomaly continued to 1 km above the ground, 5,041 points, error 5% + 10 nT; second-order trends removed
MODELS	RUNS
Density contrast (−0.2 to +0.5 g/cc) and susceptibility (0 to 1 SI) on 335,518 cells below the ground (1 km × 1 km × 250 m); each sparse with $\alpha_s = 1$ and depth weighting $\beta = 1$	7 joint inversions on AWS EC2 (ap-south-1, c5.9xlarge and c5.18xlarge), 22–68 minutes each, \$10.36 in total; the same couplings on a 2 km mesh locally

Summary

- **Without a coupling the joint inversion returns the two single inversions** (cell-by-cell correlation 0.97 and 0.94). The two uncoupled models have little in common cell by cell: 21% of the magnetic cells are dense or light, and 30% of the susceptibility model's edges lie on an edge of the density model.
- **The two structural couplings act on the susceptibility model and leave the density model alone** (correlation with the uncoupled density 0.92 and 0.96; susceptibility 0.58 and 0.69). The cross-gradient makes the gradients parallel (measure 0.43 → 0.07), but mostly by keeping the edges of the two models apart (2% shared); the joint total variation puts more of them in the same cells (44%) at some cost of fit (gravity χ^2/N 1.26).
- **The linear correspondence forces one relation on the belt and spoils the density model:** the dense body is pulled up to the magnetic rock (0.0–3.5 km instead of 3.0–7.5 km under the main high) and 46% of the density model ends below the core, while both datasets are still fitted (χ^2/N 1.12 and 1.12).
- **PGI gives models made of the rock units it is told about** (69% of the magnetic cells dense or light; χ^2/N 1.10 and 1.18), but spreads them: only 46% and 36% of the two models lie in the core. It is the slowest of the SimPEG couplings and was stopped after 43 of 60 iterations; its units are our assumptions.
- **The group lasso is not ready for field data of this size.** Without bounds it put 21.9 g/cc into padding cells on the 2 km mesh; bounds were added to the solver for this test. With bounds it couples as intended (96% of the magnetic cells anomalous in density, 35% in its control), but 69% of its susceptibility model lies below the core, and the full-resolution run needed an hour on 72 cores for 9 of its 13 values of λ_1 , where we stopped it. On the 2 km mesh, run to the end, it overfits gravity (χ^2/N 0.40) and underfits magnetics (2.86).

- **For this belt the structural assumption is the right one.** The magnetic rock (iron formation) lies on the margins of the dense rock (the greenstone pile), so the two models share boundaries, not values: a coupling that ties the values together, by one relation or by one support, moves one model to where the other is. The data fit does not tell the couplings apart, so the choice has to come from the geology.
- **No coupling fixes the depth.** The depth of each model still follows its own regularization (β); a coupling moves one model towards the other, which helps only where they really are the same rock.

1 Set-up

The two datasets are those of the single-method reports: the Bouguer anomaly with the terrain correction at 1 km, and the magnetic anomaly continued upwards to 1 km above the ground. They share one mesh that follows the ground ($84 \times 84 \times 50$ cells, 335,518 below the ground), so that a cell of the density model and a cell of the susceptibility model are the same rock.

In every run each model is regularized as in the reference run of its own report: sparse (IRLS) with $\alpha_s = 1$, $p = [0, 2, 2]$ and Li & Oldenburg depth weighting with $\beta = 1$, within its bounds. One trade-off parameter serves both models; the pipeline balances the two regularizations against their data and cools it until $\chi^2 = N$ for the 10,081 data together, with at most 60 iterations and 40 IRLS iterations. PGI and the group lasso replace this regularization by their own. The coupling weight is 1 in the pipeline's unit-free scale (as strong as the regularization).

Coupling	Kind	What it assumes	Iterations	EC2 machine	Time (min)
No coupling	—	Nothing: the two models only share the mesh.	44	c5.9xlarge	22
Cross-gradient	structural	The gradients of the two models are parallel, or one of them is zero (Gallardo & Meju 2003).	45	c5.9xlarge	26
Joint total variation	structural	The two models change in the same places (Haber & Holtzman Gazit 2013).	44	c5.9xlarge	28
Linear correspondence	petrophysical	One linear relation in every cell: density contrast = $0.5 \times$ susceptibility, the ratio of the two upper bounds.	52	c5.9xlarge	26
PGI (rock units) †	petrophysical	Every cell belongs to one of a few rock units with a density and a susceptibility (Astic & Oldenburg 2019).	43	c5.9xlarge	68
Group lasso †	joint sparsity	Few cells are anomalous, and a cell is anomalous in both models or in neither (Utsugi 2025).	9	c5.18xlarge	59
L1 + L2 by ADMM, no coupling (the group lasso's control) †	control	Few cells are anomalous in each model, no pairing: what the group lasso's solver does without the coupling.	8	c5.18xlarge	59

† Stopped by us before the end, to limit the cost: PGI (rock units) after 43 of 60 iterations; Group lasso after 9 of 13 values of λ_1 ; L1 + L2 by ADMM, no coupling (the group lasso's control) after 8 of 13 values of λ_1 . Their models are intermediate.

The couplings are those of `geoinv3d/methods/coupling.py` (docs/joint_couplings.md). PGI's units: iron formation (+0.5 g/cc, 0.5 SI), greenstone (+0.3 g/cc, 0.005 SI), light granite (−0.15 g/cc, 0.002 SI) and the background; the densities follow the rock samples, the susceptibility of the iron formation is what the magnetic data need in a 1 km cell, not a measurement. The group lasso uses the settings of its paper (datasets balanced by their largest amplitudes, λ_1 from the L-curve over 13 values, $\lambda_2 = 0.3$); it had no bounds before this test, and they were added for it (Section 4).

2 Results

2.1 Data fit

Coupling	Gravity χ^2/N	RMS (mGal)	Max residual (mGal)	Magnetic χ^2/N	RMS (nT)	Max residual (nT)
Single inversions (same settings)	0.93			0.90		
No coupling	1.06	0.51	4.7	0.91	50	659
Cross-gradient	0.96	0.49	4.6	0.91	54	924
Joint total variation	1.26	0.56	5.4	1.13	56	833
Linear correspondence	1.12	0.53	4.7	1.12	44	581
PGI (rock units) †	1.10	0.52	2.8	1.18	49	665
Group lasso †	2.07	0.72	6.1	6.86	61	625
L1 + L2 by ADMM, no coupling (the group lasso's control) †	4.21	1.03	7.6	12.18	86	805

Red: χ^2/N outside 0.7–1.3. The first row gives the single inversions with the same settings. † stopped early (Section 1).

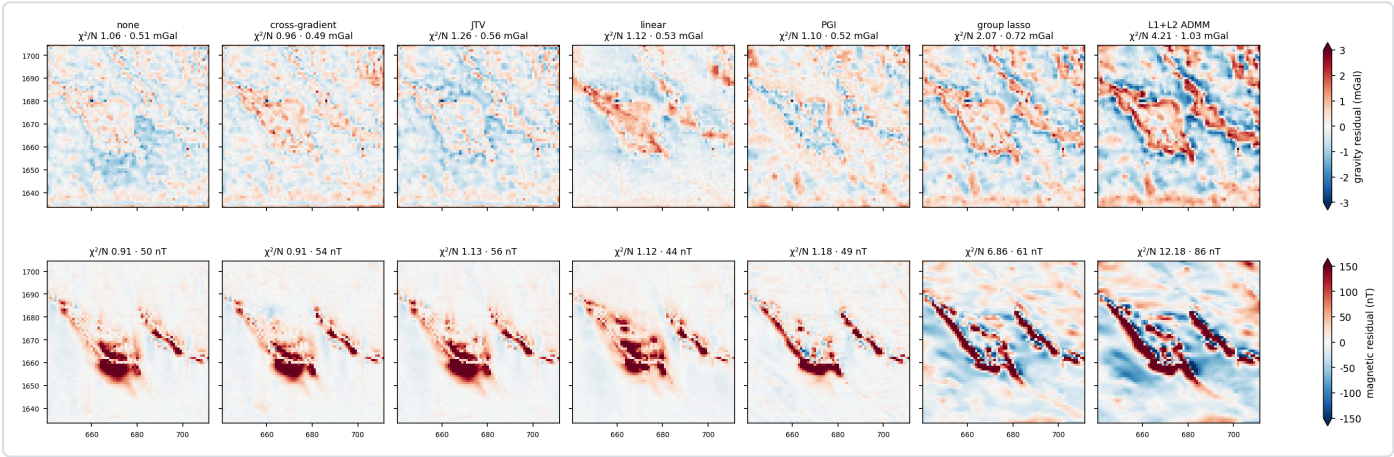


Figure 1 Residuals (observed – predicted) of the gravity data (top, ± 3 mGal) and the magnetic data (bottom, ± 150 nT) for each coupling.

Inverted together without a coupling, the two models are those of the single inversions (cell-by-cell correlation 0.97 for density and 0.94 for susceptibility): the joint set-up itself changes nothing. The residual of the magnetic data south of the Sandur belt, described in the magnetic report, is there in every run: no coupling repairs what the induced-magnetization model cannot explain.

2.2 What each coupling does to the models

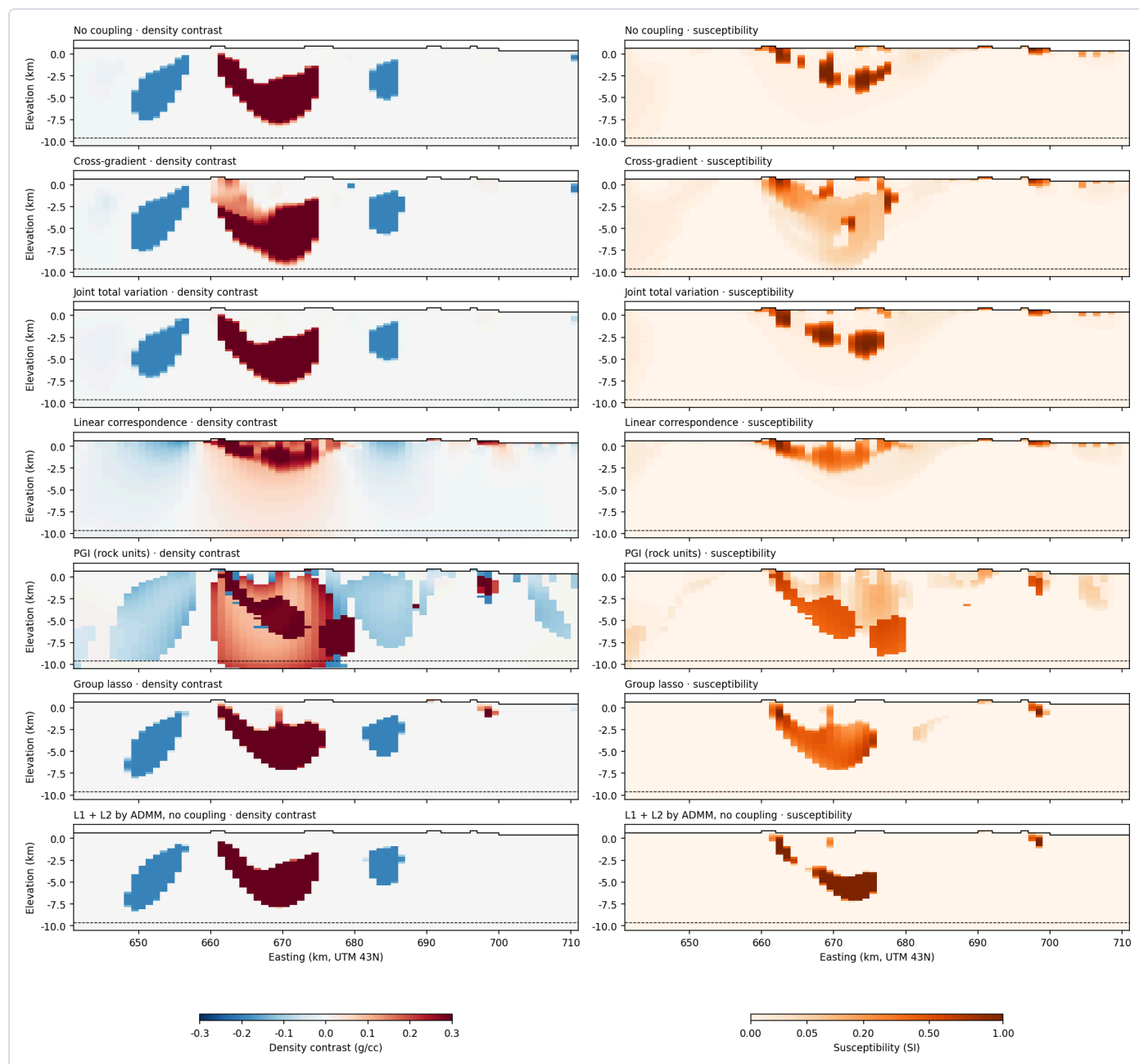


Figure 2 E–W sections through the Sandur belt (northing 1667.5 km): density contrast (left, ± 0.3 g/cc) and susceptibility (right, square-root scale to 1 SI) for each coupling. The black line is the ground of the mesh, the dashed line the base of the core.

Coupling	Density: core	below the core	main high, half-max km	centroid km	correlation with the uncoupled model	Susceptibility: core	below the core	Sandur belt, centroid km	SI·km ³	correlation with the uncoupled model
No coupling	89%	9%	3.0–7.5	5.3	1.00	70%	10%	2.0	151	1.00
Cross-gradient	90%	9%	2.5–8.5	5.7	0.92	71%	8%	2.7	165	0.58
Joint total variation	87%	10%	3.0–7.5	5.2	0.96	74%	10%	2.2	148	0.69
Linear correspondence	41%	46%	0.0–3.5	5.0	0.36	58%	19%	1.4	151	0.49
PGI (rock units) †	46%	22%	0.8–12.3	6.3	0.36	36%	22%	4.1	223	0.22

Coupling	Density: core	below the core	main high, half-max km	centroid km	correlation with the uncoupled model	Susceptibility: core	below the core	Sandur belt, centroid km	SI·km ³	correlation with the uncoupled model
Group lasso †	45%	18%	2.8–6.8	4.8	0.74	19%	69%	4.0	244	0.15
L1 + L2 by ADMM, no coupling (the group lasso's control) †	45%	8%	3.0–7.8	5.4	0.80	13%	79%	4.4	207	0.10

Depths below the ground. Main high: the density profile under the main Bouguer high (671.5, 1664.5 km). Sandur belt: susceptibility × volume in the box of the magnetic report. Correlations are cell by cell in the core columns. Red: less than half in the core, or more than 30% below it.

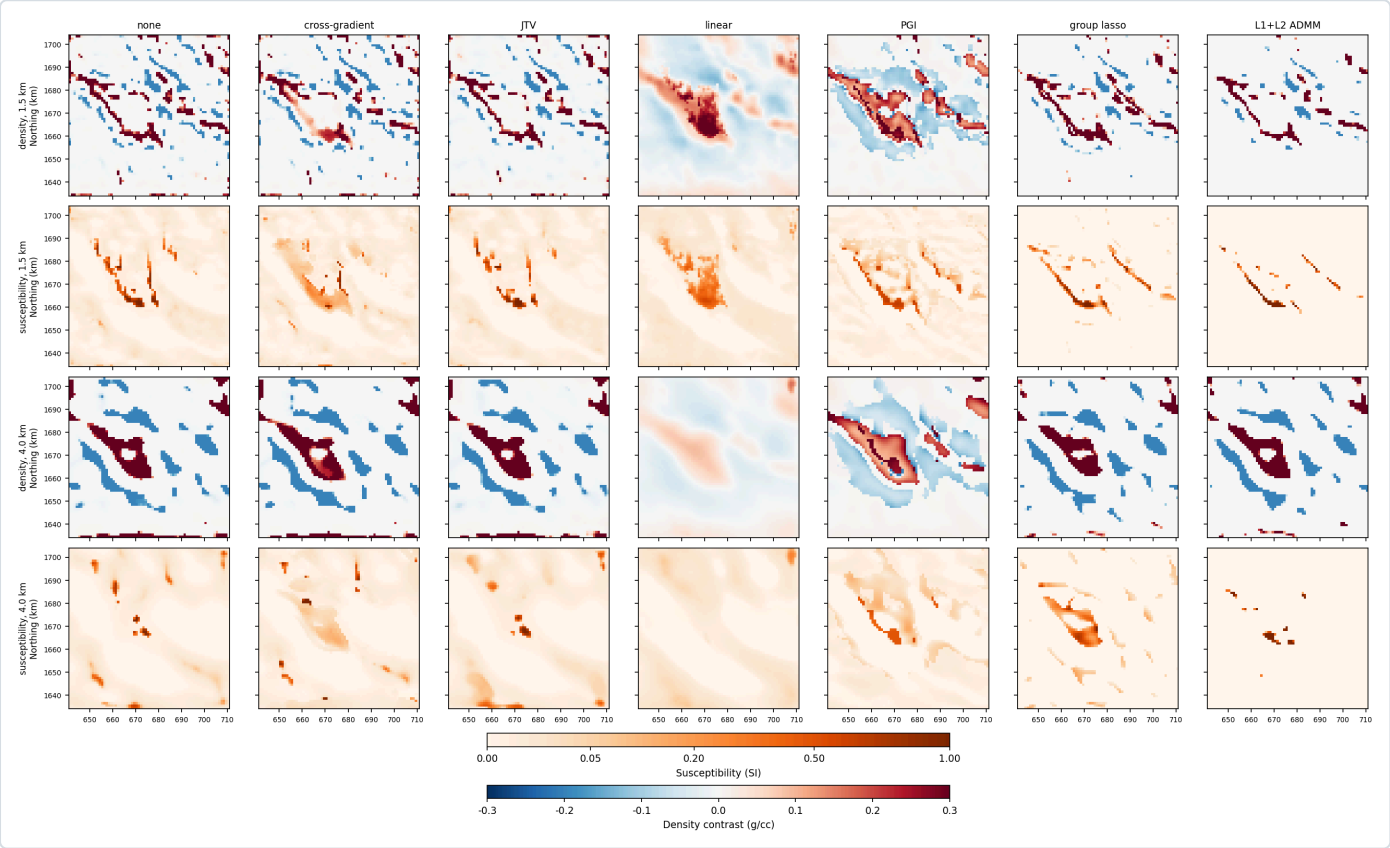


Figure 3 Layers at about 1.5 km and 4 km below the mean ground: density contrast and susceptibility for each coupling (core area only).

- Cross-gradient.** It does what it is built for: the cross-gradient measure of Section 2.3 falls from 0.43 to 0.07, with both datasets fitted (χ^2/N 0.96 and 0.91). But it gets there mostly by the second way of making a cross product zero. The susceptibility model gives up its sharp sheets for a smooth shell that follows the upper boundary of the dense body, and only 2% of its edges now lie on an edge of the density model (30% uncoupled): where one model has an edge, the other is flat. The density model hardly changes (correlation 0.92 with the uncoupled one); the susceptibility model changes a great deal (0.58), and its centroid in the belt moves from 2.0 to 2.7 km, towards the dense body.
- Joint total variation.** It rewards edges in the same cells, and the share of susceptibility edges on a density edge rises from 30% to 44%; the directions of the gradients are not constrained (measure 0.43). The magnetic rock is gathered into fewer, larger blocks inside the dense body; the density model stays as it was (0.96). It costs data fit: χ^2/N 1.26 for gravity and 1.13 for magnetics after 44 iterations.
- Linear correspondence.** The two models become one: 94% of the magnetic cells are anomalous in density too, and |density| correlates with susceptibility at 0.71. The relation is wrong for this belt and the density model pays

for it: the dense body is pulled up to the magnetic rock (half-maximum range 0.0–3.5 km under the main high, against 3.0–7.5 km), and what a shallow body cannot explain is put below the core (46% of the density model, against 9%). Both datasets are still fitted, to χ^2/N 1.12 and 1.12: the fit does not warn.

- **PGI.** The run was stopped by us after 43 of 60 iterations (68 minutes), before PGI had finished its schedule, so this is an intermediate model. With the three rock units given, 69% of the magnetic cells are dense or light, and the cross-plot (Section 2.3) shows the clusters of the units. χ^2/N is 1.10 for gravity and 1.18 for magnetics. The models are much less compact than the sparse ones: only 46% of the density model and 36% of the susceptibility model lie in the core, the dense body under the main high spans 0.8–12.3 km, and the magnetic rock of the belt is centred at 4.1 km, 2.1 km deeper than uncoupled. PGI weights its cells by their sensitivity, the weighting that filled the bottom of the mesh in the single magnetic inversions, not by the depth weighting ($\beta = 1$) of the other runs; and its units were chosen by us from the rock samples and the single inversions, not measured.
- **Group lasso.** Stopped by us at the 9th of 13 values of λ_1 , after an hour on 72 cores: the model shown is the one of the smallest λ_1 reached, and it does not fit the data yet (χ^2/N 2.1 for gravity, 6.9 for magnetics). Its coupling is visible all the same: 96% of the magnetic cells are anomalous in density too, against 35% in its control without the pairing, and $|\text{density}|$ correlates with susceptibility at 0.56. But 69% of the susceptibility model lies below the core and 55% of the density model outside it: its weighting does not keep the model where the data resolve it (Section 5).

2.3 How alike the two models are

Measured in the core of the mesh. The cross-gradient measure is the sum of $|\nabla\rho \times \nabla\chi|^2$ over the sum of $|\nabla\rho|^2|\nabla\chi|^2$: 0 when the gradients are parallel wherever both exist, 1 when they are perpendicular. An edge is a cell whose squared gradient exceeds 1% of the model's largest; a cell is anomalous when its value exceeds a tenth of the model's largest.

Coupling	Cross-gradient (0 = parallel gradients)	Susceptibility edges on a density edge	Magnetic cells that are dense or light	Cell-by-cell correlation of $ \text{density} $ and susceptibility	Correlation of the integrated maps
No coupling	0.43	30%	21%	0.06	0.25
Cross-gradient	0.07	2%	30%	0.06	0.25
Joint total variation	0.43	44%	36%	0.13	0.31
Linear correspondence	0.02	92%	94%	0.71	0.58
PGI (rock units) †	0.04	66%	69%	0.58	0.39
Group lasso †	0.04	81%	96%	0.56	0.11
L1 + L2 by ADMM, no coupling (the group lasso's control) †	0.42	31%	35%	0.14	0.15

Green: clearly more alike than the uncoupled models (the first row).

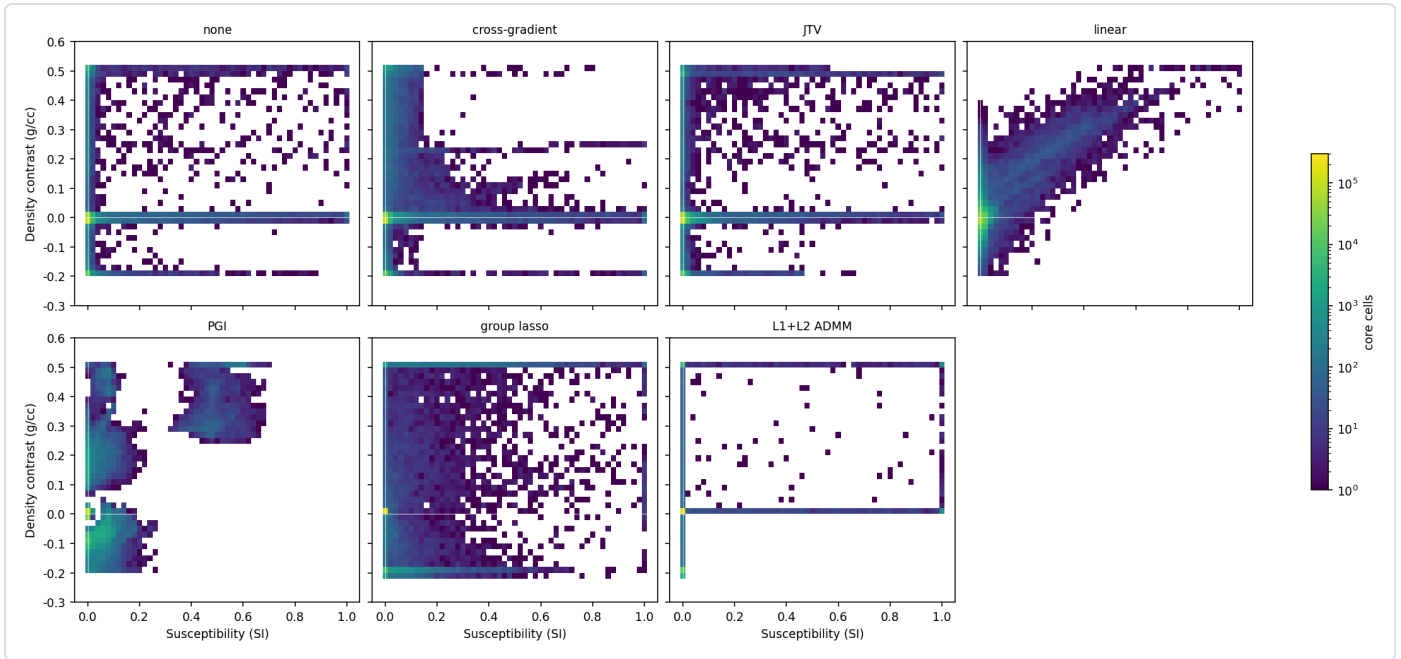


Figure 4 Density contrast against susceptibility, cell by cell in the core (logarithmic counts). Each coupling leaves its own signature.

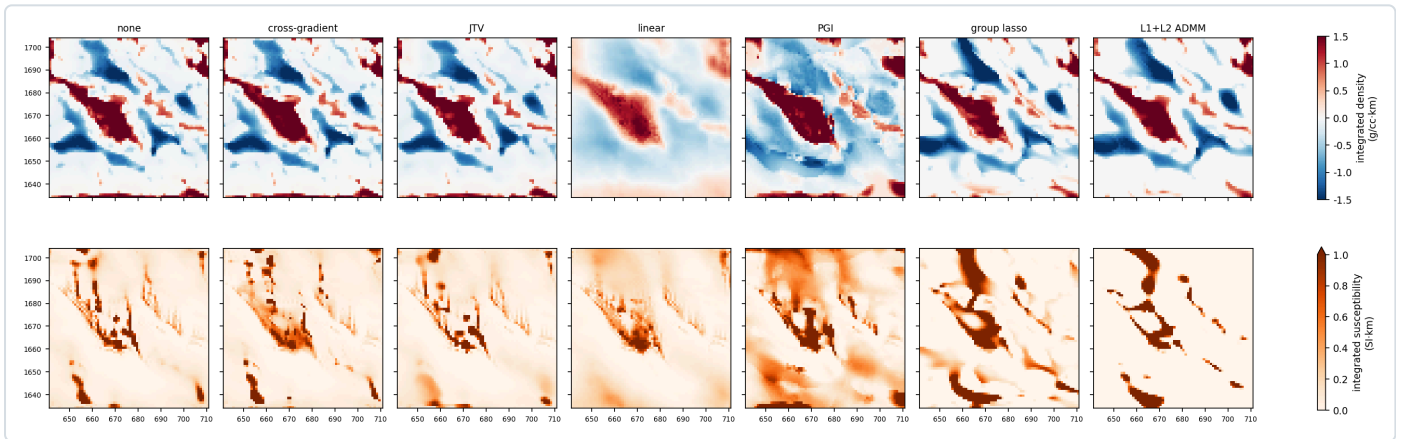


Figure 5 Vertically integrated density (top) and susceptibility (bottom) for each coupling.

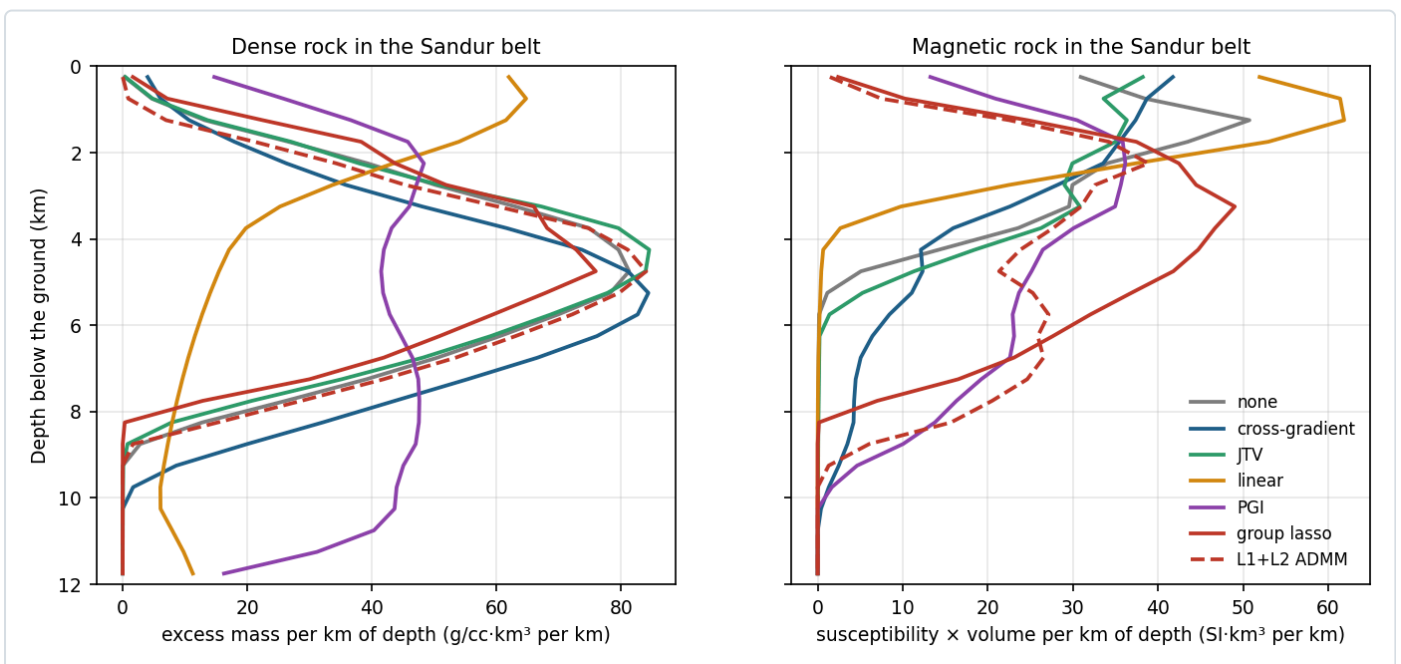


Figure 6 Dense rock (left: positive density contrast × volume) and magnetic rock (right: susceptibility × volume) in the Sandur belt, per kilometre of depth below the ground, for each coupling.

3 Which coupling suits this belt

The single-method reports showed what the two datasets see in the Sandur belt: a thick body of dense, weakly magnetic rock (the greenstone pile; amphibolite and metabasalt samples of 2.9–3.0 g/cc and 0.0002–0.004 SI), and thin sheets of strongly magnetic rock along its margins and its south-eastern closure (the banded iron formation), which are dense too (3.4 g/cc) but too thin to dominate the gravity. A coupling is an assumption about how the two properties are related; on this belt the assumptions fare as follows.

- **Same boundaries (structural couplings): consistent with the geology.** The iron formation bounds the greenstone, so edges of the susceptibility model should lie on edges of the density model. The joint total variation asks for exactly that and delivers it in part; the cross-gradient asks only for parallel gradients and is satisfied, more cheaply, by a smooth susceptibility model whose edges avoid those of the density model. Both leave the density model unchanged: the gravity data and their regularization determine it, and the magnetic sheets are too thin to reshape it.
- **One relation between the values (linear correspondence): contradicted.** The densest rock here is not the most magnetic. Forcing density = 0.5 × susceptibility moves the dense body into the magnetic sheets and leaves the rest of the gravity anomaly to the padding. A relation fitted to samples would fail in the same way, because two rock types with different relations make up the belt.
- **A few rock units (PGI): the right idea, given the right units.** Two units, a dense magnetic one and a dense non-magnetic one, describe this belt, and PGI is the only coupling that can hold both. It needs their values: here the susceptibility of the iron formation at the scale of a cell is unknown (0.05 SI in the hand samples, 0.3–1 SI in the inversions), so the run shows the consequences of our choice. It also needs the depth weighting of the other runs, in place of its sensitivity weighting, to keep the units in the core.
- **The same cells are anomalous (group lasso): partly true.** The iron formation is both dense and magnetic, the greenstone is dense only. The group lasso lets a cell be anomalous in one property and not the other, at a price, so it does not force the greenstone to be magnetic; but it pulls the magnetic rock into the dense cells (96% of the magnetic cells are anomalous in density, against 21% uncoupled). On the synthetic test of 30 September it was the best coupling because there every magnetic body was a dense body; here most of the dense rock is not magnetic.

The common point: a coupling improves a model where its assumption holds and damages it where it does not, and the data fit says little about which is the case. The measures of Section 2.3 say what a coupling did, not whether it was right.

4 The 2 km study, and bounds for the group lasso

Before the EC2 runs every coupling was run locally on a 2 km × 2 km × 500 m mesh (1,296 + 1,296 data):

Run (2 km mesh, local)	Gravity χ^2/N	Magnetic χ^2/N	Density range (g/cc)	Susceptibility range (SI)	Density in the core	Susceptibility in the core	Time (min)
No coupling	0.93	0.81	−0.20 to 0.50	0.00 to 1.00	88%	62%	7
Cross-gradient	0.84	0.85	−0.20 to 0.50	0.00 to 1.00	89%	58%	20
Joint total variation	0.92	0.97	−0.20 to 0.50	0.00 to 1.00	88%	68%	16
Linear correspondence	1.13	1.04	−0.20 to 0.50	0.00 to 0.99	53%	53%	16
PGI (rock units)	1.08	1.00	−0.20 to 0.50	0.00 to 0.99	58%	33%	49

Run (2 km mesh, local)	Gravity χ^2/N	Magnetic χ^2/N	Density range (g/cc)	Susceptibility range (SI)	Density in the core	Susceptibility in the core	Time (min)
Group lasso without bounds	0.39	1.72	-0.67 to 21.92	-0.64 to 0.64	22%	52%	24
Group lasso	0.40	2.86	-0.20 to 0.50	0.00 to 1.00	35%	17%	9
L1 + L2 by ADMM, no coupling (the group lasso's control)	0.41	3.23	-0.20 to 0.50	0.00 to 1.00	42%	13%	9
Group lasso, error-weighted	0.38	2.62	-0.20 to 0.50	0.00 to 1.00	31%	17%	9
L1 + L2 by ADMM, error- weighted, no coupling	0.41	2.79	-0.20 to 0.50	0.00 to 1.00	42%	14%	23

Red: χ^2/N outside 0.7–1.3, or values outside the bounds of the other runs.

- **The coarse runs behave like the full-resolution ones** for the four SimPEG couplings that finished at both resolutions: the cross-gradient lowers the cross-gradient measure, the joint total variation raises the share of common edges, the linear correspondence ties the values. On a laptop the uncoupled run took 7 minutes, the cross-gradient 20 and PGI 49: enough to choose a coupling before paying for the full runs.
- **PGI on the coarse mesh ran to the end** (60 iterations, χ^2/N 1.08 and 1.00); like the full-resolution run it leaves much of the susceptibility model outside the core (33% inside).
- **Without bounds the group lasso is not usable on these data.** Its first version had none, and on the 2 km mesh it returned density contrasts from -0.67 to 21.9 g/cc and negative susceptibilities (to -0.64 SI), with 78% of the density model outside the core: the large padding cells, which the data barely see, take whatever values fit. On the synthetic tests the padding was small and this did not show.
- **Bounds were added to the solver** (`GroupLassoProblem(bounds=...)`, used by the pipeline with each model's bounds). In ADMM's shrinkage step a sign constraint, such as susceptibility ≥ 0 , is projected before the group shrink, which is the exact proximal step; other bounds are clipped after it. The group lasso runs of this report use the bounds of the other couplings.
- **With bounds, two problems remain.** The fit is unbalanced: gravity χ^2/N 0.40, magnetics 2.86 (the paper's balance of the datasets by their largest amplitudes, λ_1 from the L-curve); weighting every datum by its error and choosing λ_1 for $\chi^2 = N$ does not repair it (0.38 and 2.62: even the smallest λ_1 of the path leaves the magnetic data underfitted). And the models leave the core: 35% of the density model and 17% of the susceptibility model are inside it. The group lasso scales every cell by the norm of its sensitivity column, which makes all cells equally easy to use; the sparse inversions of this report needed a much weaker depth weighting ($\beta = 1$) to stay in the core.

5 Problems found, and what is unfinished

This test was the first use of the joint inversion on field data at this size. It was stopped before every run had finished; the table lists what went wrong, what was fixed on the way and what is left.

Problem	What happened	State
Group lasso: no bounds	Unphysical values in the padding (21.9 g/cc, negative susceptibility) on the 2 km mesh.	Fixed: bounds in the ADMM solver and in the pipeline; tests added (TestBounds).
Group lasso: unbalanced fit	Gravity overfitted, magnetics underfitted, with the paper's amplitude scaling and with error weighting (2 km mesh).	Open. Needs a separate weight or target per dataset (as the SimPEG path has), or a wider λ_1 path.
Group lasso: model outside the core	More than 80% of the susceptibility model outside the core, at both resolutions; the sensitivity weighting ($\gamma = 2$) has no preference for the cells the data resolve.	Open. Tests of $\gamma = 0.5, 1$ and 1.5 on the 2 km mesh were started and stopped unfinished; the settings are in <code>joint_params.py</code> .

Problem	What happened	State
Group lasso: cost at full resolution	28 minutes of set-up and about 5 minutes per value of λ_1 on 72 cores; stopped at the 9th of 13 values. Two further runs with error weighting were stopped after 35 minutes, at the 1st and 2nd value, and are not reported.	Open. A shorter λ_1 path (or a fixed λ_1 from the 2 km run) and the error weighting would make it affordable.
PGI: slow, and not compact	1.4–1.6 minutes per iteration on 36 cores; stopped after 43 of 60 iterations. Half of both models outside the core.	Open. Run it to the end; weight it by depth ($\beta = 1$) instead of by sensitivity; get measured units.
Joint total variation: fit	Gravity χ^2/N 1.26 after 44 iterations: the target was not reached within the iteration limit.	Open. More iterations or a smaller coupling weight.
Magnetic data: systematic residual	The high south of the Sandur belt is underfitted in every single and joint run.	Open. Points to remanent magnetization; needs a magnetization-vector inversion.
EC2: instances lost at launch	With eight launches at once, four instances were reported as terminated seconds after launch and left running untracked.	Fixed in <code>c1oud/ec2.py</code> (a just-launched instance unknown to <code>describe_instances</code> is 'launching'); the four were terminated by hand and rerun.
Stations inside the ground	With the ground as a staircase of cells, 1,900 of 5,040 gravity stations lay inside the top cell of their column.	Fixed: such stations are moved to the top of their column (72 m on average).

AWS cost of the three reports (gravity with terrain, magnetics, joint): about \$17 on EC2, of which \$6 for the two group lasso runs of this report and \$3.6 for two that were stopped without a result.

6 Recommendations

- **Choose the coupling from the geology, not from the fit.** Same rock in both models (one body, dense and magnetic): group lasso once it is fixed, or PGI with measured units. Different rock sharing boundaries, as in this belt: a structural coupling. Unknown: invert separately first and compare the two models, as in Section 5 of the magnetic report.
- **For this area: the joint total variation or the cross-gradient**, read together with the uncoupled models. Report what is common to them; do not read the coupled susceptibility model as better resolved.
- **Do not use the linear correspondence** unless one rock type carries both anomalies.
- **PGI needs petrophysics and a depth weighting:** susceptibility measurements of the fresh iron formation and of the greenstone at the scale of a cell (logs, not hand samples) would turn its units from an assumption into a constraint.
- **Group lasso: finish it on the 2 km mesh before running it at full resolution again.** Bounds are in; the balance between the two datasets and the weighting that keeps the model in the core are not (Section 5).
- **Depth still needs independent information.** The couplings do not supply it; the mapped dips of the iron formation, boreholes, or a magnetization-vector inversion for the remanence would.

Files

- Parameters of every run: `examples/output/karnataka_joint/scripts/joint_params.py` ; launched with `deploy/ec2_multi_run.py karnataka-joint` and `scripts/run_lowres.py`
- This report and its figures: `scripts/make_figures.py` , `scripts/build_report.py` , `scripts/report_text.py` ; results in `data/ec2_runs/` and `data/lowres_runs/`
- All runs in one interactive workflow (DAG viewer): `karnataka_joint.geoinv3d_viewer.html`
- The single-method reports: `examples/output/karnataka_gravity_terrain/` , `examples/output/karnataka_magnetic/`

